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Review – Prostate Cancer

Comparing the Performance of Digital Rectal Examination and Prostate-specific Antigen as a Screening Test for Prostate Cancer: A Systematic Review and Meta-analysis

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Abstract

Background and objective: Although digital rectal examination (DRE) is recommended in combination with prostate-specific antigen (PSA) for detection of prostate cancer (PCa), there are limited data to support its use as a screening/early detection test. Our objective was to assess the diagnostic value of DRE in screening for early detection of PCa.

Methods: In August 2023, we queried the PubMed, Scopus, and Web of Science databases to identify prospective studies simultaneously investigating the diagnostic performance of DRE and PSA for PCa screening. The primary endpoints were the positive predictive value (PPV) and cancer detection rate (CDR) of DRE. Secondary endpoints included the PPV and CDR of both PSA alone and in combination with DRE. We conducted meta-regression analysis to compare the CDR and PPV of different screening strategies. This meta-analysis is registered on PROSPERO (CRD42023446940).

Key findings and limitations: We identified eight studies involving 85,798 participants, of which three were randomized controlled trials and five were prospective diagnostic studies, that reported the PPV and CDR of both DRE and PSA for the same cohort. Our analysis revealed a pooled PPV of 0.21 (95% confidence interval [CI] 0.13–0.33) for DRE, which is similar to the PPV of PSA (0.22, 95% CI 0.15–0.30; $p = 0.9$), with no benefit

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from combining DRE and PSA (PPV 0.19, 95% CI 0.13–0.26; $p = 0.5$). However, the CDR of DRE (0.01, 95% CI: 0.01–0.02) was significantly lower than that of PSA (0.03, 95% CI 0.02–0.03; $p < 0.05$) and the combination of DRE and PSA (0.03, 95% CI 0.02–0.04; $p < 0.05$). The screening strategy combining DRE and PSA was not different to that of PSA alone in terms of CDR ($p = 0.5$) and PPV ($p = 0.5$).

Conclusions and clinical implications: Our comprehensive review and meta-analysis indicates that both as an independent test and as a supplementary measure to PSA for PCa detection, DRE exhibits a notably low diagnostic value. The collective findings from the included studies suggest that, in the absence of clinical symptoms and signs, DRE could be potentially omitted from PCa screening and early detection strategies.

Patient summary: Our review shows that the screening performance of digital rectal examination for detection of prostate cancer is not particularly impressive, suggesting that it might not be necessary to conduct this examination routinely.

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1. Introduction

The age-standardized incidence rate of prostate cancer (PCa) has been gradually increasing globally, and it is ranked as the most common cancer among men in Europe and the second most common in the USA [1]. This increase may be attributed in part to the widespread adoption of prostate-specific antigen (PSA) testing for PCa screening/early detection. The ERSPC trial demonstrated that the use of PSA screening tests was associated with a reduction in cancer-specific mortality rates [2]. Thus, PSA is widely used as a screening/early detection test for PCa together with digital rectal examination (DRE). DRE was used before the introduction of PSA to help in identifying palpable suspicious areas on the prostate and its size (eg, prostate volume). With the advent of PSA as a screening/early detection biomarker, the value of DRE for cancer detection has steadily decreased. Results from the PLCO cancer screening trial indicate that clinically significant PCa was detected in only 2% of cases with suspicious findings on DRE despite a PSA level below the cutoff [3]. In primary detection, the diagnostic performance of DRE tends to be moderate to poor. Furthermore, there is significant variability in the interpretation of DRE findings among clinicians [4–6]. DRE continues to be used together with PSA for PCa detection strategies in guidelines [7,8]. Considering these findings, DRE could potentially be omitted from screening/early detection strategies, especially as the incorporation of magnetic resonance imaging (MRI) has the potential to enhance screening/early detection protocols [9,10]. In addition, DRE is an invasive examination that men usually dislike and is a reason for avoiding visits to their health care provider, which can lead to a delay in PCa diagnosis and other health impacts. Therefore, we performed a systematic review and meta-analysis to address the diagnostic role of DRE in screening/early detection of PCa. We also assessed the diagnostic performance of DRE in comparison to PSA and to a combination of PSA and DRE in PCa screening strategies.

2. Methods

Our study protocol is registered with the International Prospective Register of Systemic Reviews database (PROSPERO: CRD42023446940). The meta-analysis adheres to the guidelines of the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement and AMSTAR2 checklist [11,12].

2.1. Study selection and characteristics

In August 2023, a comprehensive search was performed using the PubMed, Scopus, and Web of Science databases to identify prospective studies that investigated the screening performance of DRE for PCa. The detailed search strategy is shown in the Supplementary material. Initial screening by article title and abstract was performed by two independent investigators to identify eligible studies. Potentially relevant studies were subjected to a full-text review. Manual searches of reference lists of relevant articles were also carried out to find additional studies of interest. Disagreements were settled via consensus with co-authors.

2.2. Inclusion and exclusion criteria

For our clinical question, we used the PICO (Population, Intervention, Comparator, Outcome) framework. Studies eligible for inclusion evaluated men undergoing screening for PCa (Population) using DRE as a screening method (Intervention) in comparison to PSA testing (Comparator) to assess the diagnostic effectiveness of these methods according to their positive predictive value (PPV) and cancer detection rate (CDR) (Outcome). Thus, the primary outcomes of interest were the PPV and CDR. Exclusion criteria were studies without original patient data, reviews, editorial comments, replies to authors, languages other than English, and studies that did not conduct both DRE and PSA testing in the same cohort. In addition, studies that enrolled participants with clinically suspected PCa, such as men presenting with elevated PSA or urinary symptoms, were excluded in favor of including studies with a community-based population, such as those composed of randomly selected individuals gathered for screening purposes.

2.3. Data extraction

Data related to study design, patient characteristics, prostate biopsy criteria, and outcomes were extracted indepen-

dently by two authors. All discrepancies were resolved via consensus with co-authors. PPV and CDR data were extracted where possible. For studies that did not provide information on PPV and CDR, these values were independently calculated by two authors using the data provided within the studies. CDR was calculated as the number of true positives divided by the total number of screened individuals [13]. In cases in which the number of true positives was divided by the number of biopsies performed, recalculation was carried out. As prostate biopsies were not performed in cases with negative screening tests, we were not able to evaluate the sensitivity, specificity, or negative predictive value.

2.4. Risk-of-bias assessment

Study quality and the risk of bias were evaluated using the ROBINS-I tool for nonrandomized studies and RoB 2 for randomized controlled trials (RCTs) as outlined in the Cochrane handbook for systematic reviews of interventions (Supplementary Fig. 1 and Supplementary Table 2) [14]. In the case of ROBINS-I, bias for every domain and overall was classi-

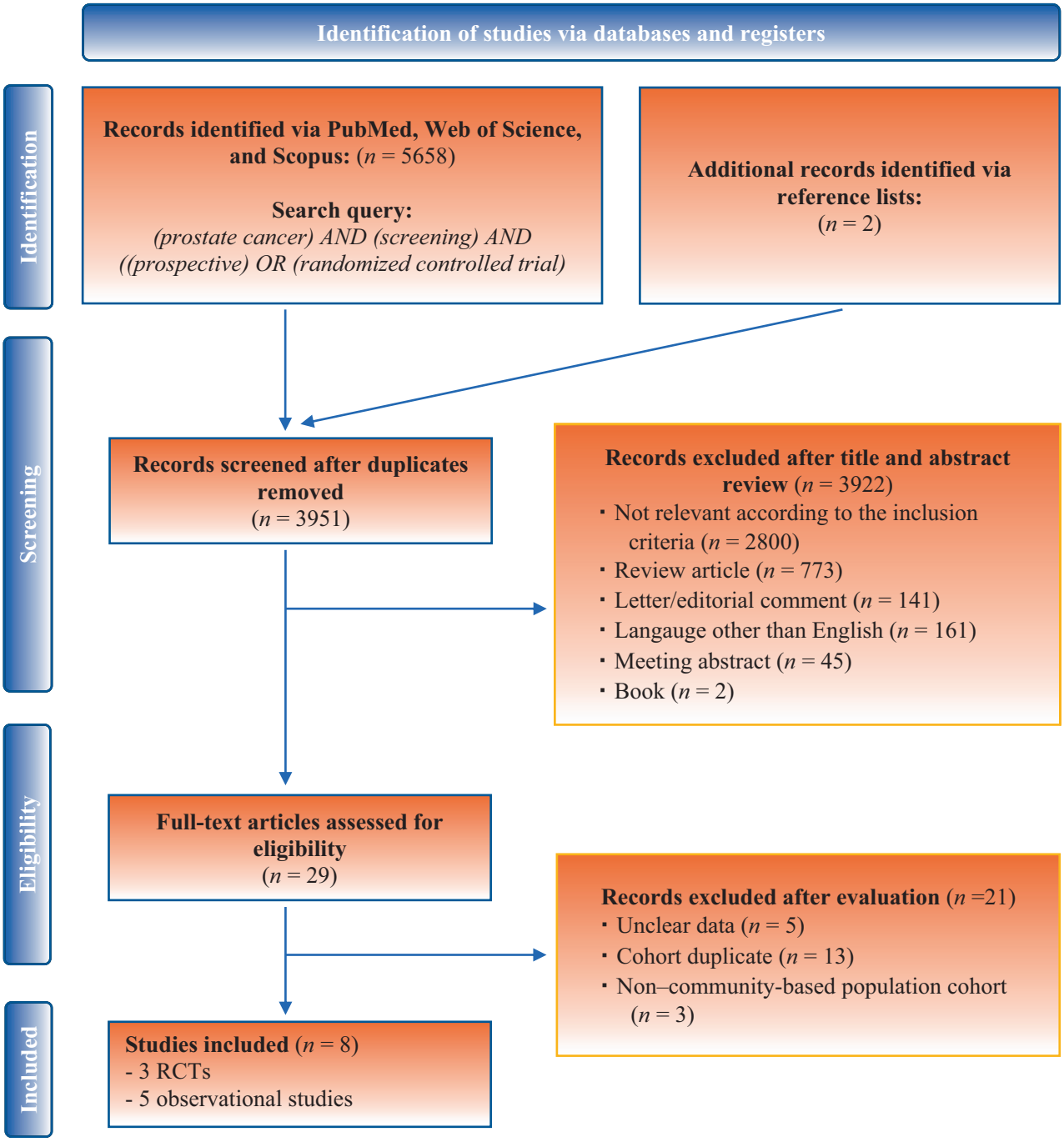


Fig. 1 – Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart detailing the article selection process. RCT = randomized controlled trial.

Table 1 – Demographics and patient characteristics in the studies included in the review

Study	Design	Participants	Definition of suspicious or abnormal DRE findings	Biopsy technique	Normal PSA range (ng/ml)	Age (yr)	PSA (ng/ml)	GS	Stage
Kirby 1994 [23] UK	PDS	Overall: 568 Screening: 568Bx: 29	NOD or ASY	TRUS Bx Random Bx + TBx	0–4	55–70	0–4.0: 483 4.0–10: 62 ≥10 : 19Not tested: 4	NA	T1: 2; T2: 4 T3: 5; T4: 0M0: 8; M1: 3
Catalona 1994 [21] USA	PDS	Overall: 6630 Screening: 6630Bx: 1167	IND, ASY, or irregularity	TRUS Bx SBx (4 cores) + TBx	0–4	50–59: 2381 (37%) 60–69: 2959 (45%) 70–79: 1161 (17%) ≥80: 129 (2%)	0–4.0: 5647 (85%) 4.1–9.9: 809 (12%) ≥10: 174 (3%)	NA	NA
Bretton 1994 [24] USA	PDS	Overall: 1027 Screening: 1027 Bx: 185	NA	TRUS or digital guided Bx	0–4	Mean 64	0–3.9: 898 4.0–9.9: 107 ≥10: 22	NA	NA
ERSPC trial Bangma 1997 [18] Netherlands	RCT	Overall: 9600 Screening: 4800Bx: 976	NA	TRUS transrectal lateral sextant Bx SBx + TBx	0–4	55–74	NA	NA	T1c: 40% T2a: 44% T2b: 8% T2c: 7% T3a: 1%T3b: 0; T3c: 0
Brett 1998 [22] Australia	PDS	Overall: 241 Screening: 211Bx: 11	NOD, IND, ASY, or AMS	NA	0–4	50–59: 88 (42%) 60–69: 68 (32%) 70–79: 55 (26%)	0–4.0: 191 (91%) >4.0: 20 (9%)	GS 5: 2GS 6: 1	NA
LUPCPD trial Candas 2000 [19] Canada	RCT	Overall: 11 811 IG: 7195 – Bx: 533 UIG: 4616– Bx: 380	NA	NA	0–3	45-80	IG 0.0–1.0: 3086 (43%) 1.1–2.0: 2153 (30%) 2.1–2.9: 838 (12%) 3.1–4.0: 427 (6%) 4.1–5.0: 212 (3%) 5.1–7.0: 218 (3%) 7.1–10.0: 120 (2%) 10.1–20.0: 98 (1%) >20.1: 43(1%)	NA	IG A2: 1 (1%) A3: 2 (1%) B0: 15 (7%) B1: 74 (36%) B2: 59 (29%) C1: 25 (12%) C2: 17 (8%) D1: 4 (2%)D2: 9 (4%)
PLCO trial Andriole 2005 [17] USA	RCT	Overall: 38 350 Screening: 34 244Bx: 1510	IND, NOD, significant ASY, or LAL	NA	0–4	55–59: 32.3% 60–64: 31.4% 65–69: 23.2%70–74: 13.2%	>4.0: 8%	GS 2–4: 10% GS 5–6: 45% GS 7: 31% GS 8–10: 12% Unknown: 2%	I/II: 83% III: 6% IV: 4%NR: 7%
Faria 2012 [22] Brazil	PDS	Screening: 17 571Bx: 1647	NA	12-sextant TRUS Bx SBx + TBX	0–4	Mean 61.2 (SD 9, range 45–98)	>4.0: 1021 (6%)	NA	NA

RCT = randomized controlled trial; PDS = prospective diagnostic study; DRE = digital rectal examination, PSA = prostate-specific antigen; GS = Gleason score; TRUS: transrectal ultrasound-guided biopsy; NA = not available; NR = not recorded; IG = invited group; UIG = uninvited group; UND = induration; NOD = nodularity; ASY = asymmetry; AMS = absence of median sulcus; Bx = biopsy; SBx = systematic BX; TBx = targeted Bx; LAL = loss of anatomic landmarks; SD = standard deviation.

fied as low, moderate, serious, or critical. The presence of confounders was determined via consensus. The ROBINS-I and RoB 2 assessments for individual studies were conducted independently by two authors.

2.5. Statistical analyses

2.5.1. Meta-analysis

Owing to the anticipated heterogeneity for outcomes among the studies included, we used a random-effects model to compute pooled PPV and CDR estimates. Forest plots for PPV and CDR were generated. We conducted a meta-regression analysis to compare DRE, PSA, and the combination of DRE and PSA,. The presence of heterogeneity for outcomes among the studies included in the meta-analysis was evaluated using Cochrane's Q test. In instances of significant heterogeneity ($p < 0.05$ for the Cochrane Q test), we tried to investigate and explain the heterogeneity [15,16]. We used funnel plots to evaluate the presence of publication bias (Supplementary Figs. 2 and 3). We did not perform the Egger test because the number of studies included was less than ten. All analyses were performed using R v4.3.0 (R Foundation for Statistical Computing, Vienna, Austria) with the *metafor* package. The statistical significance level was set at $p < 0.05$.

3. Results

3.1. Study selection and characteristics

The search string is presented in Figure 1. According to our inclusion criteria, a total of eight studies were finally selected (n = 85 798 patients). We identified three RCTs involving 59 761 patients [17–19] and five prospective diagnostic studies involving 26 037 patients [20–24]. The patient characteristics and outcomes across the studies are shown in Tables 1 and 2. The age of participants ranged from 45 to 98 yr. The definition of abnormal or suspicious DRE findings varied among the studies, as shown in Table 1. Findings reported in the studies included nodularity, induration, asymmetry, and loss of anatomic landmarks [17,20,21,23]. However, specific details regarding abnormal or suspicious DRE findings were not provided in LUPCPD, ESRPC, and the study by Faria et al. [18,19,22,24]. Most studies used a PSA cutoff of ≥4.0 ng/ml, except for one study that used ≥3.0 ng/ml [19]. In all studies, prostate biopsies were recommended when either DRE or/and PSA were abnormal. Among all the participants, 72 246 men underwent screening tests. Of these, 4718 (6.6%) men had suspicious or abnormal DRE findings and 6328 (8.8%) had elevated PSA. In total, 9473 (13.1%) men had suspicious or abnormal findings for either DRE or PSA, and 6058 (8.4%) underwent prostate biopsy.

3.2. Risk of bias

Results for the risk of bias for each domain across the studies are presented in Supplementary Figure 1 and Supplementary Table 2. Although one RCT had bias concerns for certain domains, the other two RCTs had low risk of bias across all domains. Although some of the prospective diagnostic studies had moderate risk for specific categories, the majority of these studies had either low or moderate overall risk of bias. Funnel plots for the analyses are presented in Supplementary Figures 2 and 3.

Table 2 – Outcomes in the studies included in the review

Study	PPV	CDR
Kirby 1994 [23]		
DRE	0.57	0.01
PSA	0.14	0.02
DRE + PSA	0.13	0.02
Catalona 1994 [21]		
DRE	0.15	0.02
PSA	0.22	0.03
DRE + PSA	0.23	0.04
Bretton 1994 [24]		
DRE	0.14	0.01
PSA	0.28	0.04
DRE + PSA	0.21	0.04
Bangma 1997 [18]		
DRE	0.27	0.02
PSA	0.29	0.03
DRE + PSA	0.23	0.04
Brett 1998 [22]		
DRE	0.11	0.01
PSA	0.11	0.01
DRE + PSA	0.1	0.01
Candas 2000 [19]		
DRE	0.02	0.01
PSA	0.17	0.03
DRE + PSA	0.15	0.03
Andriole 2005 [17]		
DRE	0.09	0.01
PSA	0.18	0.01
DRE + PSA	0.12	0.02
Faria 2012 [22]		
DRE	0.36	0.01
PSA	0.47	0.04
DRE + PSA	0.42	0.03

DRE = digital rectal examination; PSA = prostate-specific antigen, PPV = positive predictive value, CDR = cancer detection rate.

3.3. Meta-analysis

3.3.1. DRE

The pooled PPV for DRE as a screening test was 0.21 (95% confidence interval [CI] 0.13–0.33) and the pooled CDR was 0.01 (95% CI 0.01–0.02; Fig. 2). There was significant heterogeneity among the studies ($p < 0.001$ for both outcomes).

3.3.2. PSA

The pooled PPV for PSA as a screening test was 0.22 (95% CI 0.15–0.30) and the pooled CDR was 0.03 (95% CI 0.02–0.03; Fig. 2). Cochrane's Q indicates significant heterogeneity among the studies included ($p < 0.001$). The PPV for the PSA test did not significantly differ from that for DRE ($p = 0.9$); however, the diagnostic performance of the PSA test was significantly better in terms of the CDR ($p < 0.01$).

3.3.3. Combination of DRE and PSA

We examined a screening strategy that combines DRE and PSA testing. The pooled PPV for the combined screening strategy was 0.19 (95% CI 0.13–0.26) and the pooled CDR was 0.03 (95% CI 0.02–0.04; Fig. 2). Cochrane's Q indicates significant heterogeneity among the studies included ($p < 0.001$). The PPV for the combined DRE and PSA pathway

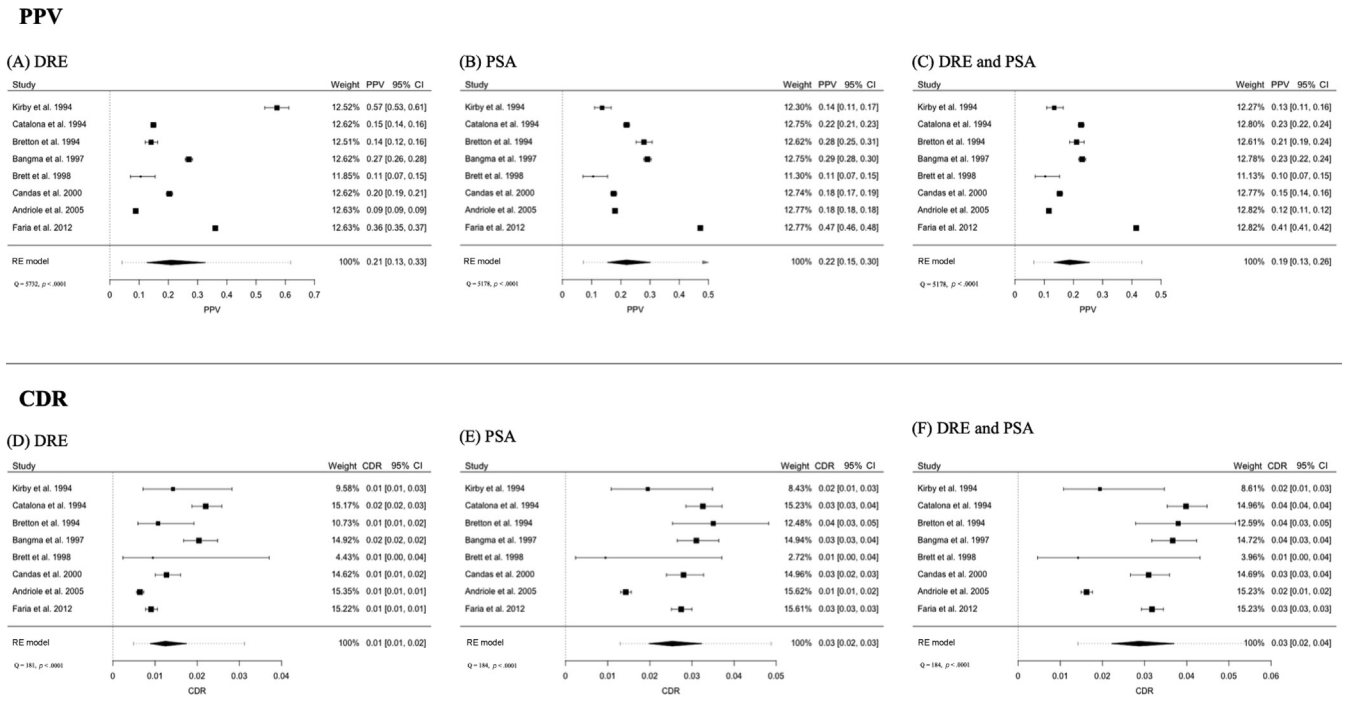


Fig. 2 – Forest plots for the PPV of (A) DRE, (B) PSA, and (C) DRE + PSA; and the CDR of (D) DRE, (E) PSA, and (F) DRE + PSA. PPV = positive predictive value; DRE = digital rectal examination; PSA = prostate-specific antigen; CDR = cancer detection rate; CI = confidence interval; RE = random effects.

was not significantly different to that for DRE alone ($p = 0.7$). Conversely, the combined strategy performed significantly better in terms of CDR ($p < 0.01$). There were no significant differences in PPV and CDR in comparison to PSA alone (both $p = 0.5$).

4. Discussion

We present the first systematic review and meta-analysis of the effectiveness of DRE for PCa screening in comparison to PSA that highlights several findings. First, our data suggest that the combination of DRE and PSA for PCa screening has no significant benefit over PSA alone in terms of PPV and CDR. Second, while our findings indicate no significant difference in PPV between DRE and PSA, PSA has a significantly higher CDR than DRE.

Although we found that DRE and PSA were similar in terms of PPV, significant differences in CDR were observed. Approximately only 20% of men with abnormal DRE or PSA have PCa, and only 1–2% of screened men had PCa identified via these screening methods. Acknowledging the risk that the high CDR for PSA screening might lead to overdiagnosis of clinically insignificant PCa, our results suggest a potential higher screening value of PSA in comparison to DRE. Data from PROBASE [25], which is an RCT targeting 45-yr-old men, demonstrated that in comparison to PSA, the true-positive detection rate of DRE was 0.22 (95% CI 0.07–0.72) and the false-positive detection rate was 2.18 (95% CI 1.50–3.17), supporting our results. In addition, according to the literature the cost of a PSA test is low; and DRE induces physical discomfort and psychological stress in patients [26,27]. Taken together, the data indicate that PSA seems to be a more effective and less distressing screening test than DRE. In fact, dislike of DRE is a frequent reason for avoiding PCa screening/early detection.

We also examined the performance of a screening strategy that combines DRE and PSA. Similar to DRE and PSA as standalone tests, approximately 20% of cases who were screened as positive had PCa. Furthermore, the combined screening pathway was only able to identify PCa in approximately 3% of screened men. There was no significant difference in PPV when the combination of DRE and PSA was compared to either DRE or PSA alone. However, there was a significant difference in CDR when comparing the combination to DRE alone, but not to PSA. On the basis of these findings, inclusion of DRE in addition to PSA does not seem to enhance the screening effectiveness. Consequently, our results suggest that PSA alone could serve as a sufficient method for PCa screening, avoiding DRE. Although DRE has historically been used as a low-cost PCa screening method [27], it has several challenges. Results from the PLCO cancer screening trial indicate that in a cohort of 5064 patients with DRE suspicion for PCa despite a PSA level below the cutoff for abnormal (eg, 4 ng/ml), only 99 (2%) were identified as having clinically significant PCa [3]. The effectiveness of DRE as a screening method is compromised not only by its limited performance but also by its highly subjective nature. The impact of this subjectivity was demonstrated by Gosselaar et al. [28], who found that the percentage of suspicious DREs varied between examiners, ranging from 4% to 28%. Additional evidence of the role of experience in this context was provided by Smith and Catalona [4]. The authors found that when DRE was conducted by experienced faculty urologists, agreement for DRE findings was as high as 94%. However, when examiners of

varying experience levels conducted DRE, the consistency in findings dropped to 82%. Considering the subjective nature and inconsistency of DRE results, its routine use in combination with PSA appears unnecessary.

While the standalone screening capability of PSA is limited, integration of MRI into the screening process is promising, especially for clinically significant PCa. In a meta-analysis conducted by Oerther et al. [29], the average PPV in grade group $\geq$2 cancer was 16% (95% CI 7–27), 59% (95% CI 39–78%), and 85% (95% CI 73–94) for lesions with a PI-RADS v2.1 score of 3, 4, and 5, respectively. Several ongoing large-scale population-based trials are currently evaluating MRI for PCa screening. The MVP study was designed to compare the PPV of PSA and MRI. Findings from this study demonstrated a significantly higher PPV for MRI (63%) than for PSA (29%; $p = 0.02$). Moreover, a significantly higher percentage of clinically significant cancers were detected in the MRI group than in the PSA group (73% vs 50%) [30]. The GÖTEBORG 2 trial, a prospective randomized population-based study in Sweden, is investigating the screening performance of the combination of PSA and MRI, with PPV of 35% and CDR of 2% reported so far [31]. On the basis of these results, it appears that the combination of MRI and PSA, rather than DRE and PSA, may offer a more effective screening strategy for PCa [29,31,32]. Certainly, a cost-efficient and more precise strategy could consist of PSA testing. When considered abnormal based on a highly sensitive cutoff, it would then trigger an MRI test, which is highly effective in ruling in clinically significant PCa and ruling out clinically insignificant PCa as well as cases with no PCa [33].

Our study has several potential limitations. First, there were variations in patient characteristics such as age, PSA, and cancer stage among the studies included in the analyses. Consequently, the cohorts may not be entirely representative of community-based populations, particularly because of varying age inclusion criteria. Furthermore, the definitions of suspicious or abnormal DRE findings, the thresholds for elevated PSA, and the biopsy method (ie, sextant biopsy, targeted biopsy) differed to some degree across the studies. Second, PPV can vary according to the prevalence of the disease. Therefore, when evaluating the effectiveness of a screening test, the sensitivity, specificity, and negative predictive value should ideally be considered. However, for ethical reasons, no study has performed prostate biopsies on individuals with negative screening results and the accurate numbers of false negatives and true negatives were unknown, making it impossible to assess these variables. Third, biopsy compliance varied among the studies, affecting the accuracy of the assessments. For instance, in the study by Kirby et al. [23], only 29 out of 82 screen-positive subjects underwent biopsies. We conducted a sensitivity analysis and subgroup analysis, but heterogeneity was present in every pattern we examined, as shown in Supplementary Figure 3. Furthermore, owing to the absence of key information such as age, PSA, and disease stage in many studies, we did not proceed with a meta-regression analysis to further investigate the sources of heterogeneity. We consider that this heterogeneity is largely influenced by the impact of these limitations mentioned. Finally, the primary goal of screening tests is to identify clinically significant PCa. However, the absence of certain data precluded us from performing analyses of the PPV and CDR specific to clinically significant PCa.

5. Conclusions

Our study demonstrates that DRE as a screening tool for PCa was similar to PSA in terms of the PPV, but with a significantly lower CDR. Furthermore, the combination of DRE and PSA did not enhance the screening efficacy in comparison to PSA alone. These findings suggest that DRE could be potentially omitted from PCa screening and early detection in the absence of clinical symptoms and signs.

Author contributions: Akihiro Matsukawa had full access to all the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis.

Study concept and design: Matsukawa, Yanagisawa, Rajwa.
Acquisition of data: Matsukawa, Yanagisawa.
Analysis and interpretation of data: Matsukawa, Yanagisawa, Rajwa.
Drafting of the manuscript: Matsukawa, Yanagisawa, Rajwa, Bekku, Kardoust Parizi, Laukhtina, Klemm, Chiujdea, Mori, Kimura, Fazekas, Miszczyk.
Critical revision of the manuscript for important intellectual content: Miki, Kimura, Karakiewicz.
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Appendix A. Supplementary data

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